Chinmay Rawat

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Research interests: how LLM agents reason and why they act as they do; evaluating that reasoning step by step against expert judgment, with metrics that measure what they claim to; and agents that reason like experts.
Open-source contributor: PyTorch · TransformerLens · marimo · OpenClaw · pylint

Education

University of Illinois Urbana-Champaign Urbana, IL
B.S. in Computer Science, Minor in Statistics; James Scholar Honors *Aug 2024 – May 2027 (expected)*

- GPA **3.80/4.00**; Dean's List (Fall 2024, Spring 2025, Fall 2025); completing the B.S. in **three years**.
- Coursework: AI, Algorithms, Databases, Compilers, Numerical Methods; in progress: NLP, System Programming.

Publications & Manuscripts

[1] **Chinmay Rawat**, Taaha Kazi, Vasu Sharma. *More Context Is Not Validation: A Baseline-First Audit of Action-Grounded LLM Listener Agents.* Social Simulation Workshop at **COLM 2026** (non-archival), poster.

[2] **Chinmay Rawat**. *When ReAct Phase Probes Look Abstract: Cue-Dependent Decodability in Gemma-2 2B and 9B.* Sole-author manuscript, submitted to **BlackboxNLP 2026**.

Research Experience

Alta Research Group, University of Illinois Urbana-Champaign Urbana, IL
Undergraduate Researcher; advisor: Prof. Hao Peng *Aug 2026 – present*

clinical-sim: an RL environment for sequential clinical diagnosis (project lead)

- Formulated diagnosis as a **POMDP**: from a one-line presentation, the agent questions, examines, and orders labs, imaging and procedures before committing to a diagnosis. Extends Microsoft's **SDBench** (which scores only the final diagnosis and its cost) to **step-level comparison with physicians** on **1,400+ real cases** transcribed from physicians reasoning aloud.
- Built in Python: an **LLM gatekeeper** that returns only the finding requested, never interprets results, and withholds answer-revealing evidence until the agent commits; a deliberative agent that logs its **belief state** each step (ranked differential with probabilities, evidence, rationale, expected result) and checks its consistency before diagnosing; SDBench's 5-point judge plus **process metrics** (coverage of physicians' actions; how its differential and confidence track theirs).
- Served **Qwen3.5-9B** with **vLLM** on NCSA Delta A40 GPUs (Slurm) alongside GPT-5.6 behind one client, with a **content-addressed cache** that makes every episode deterministic, resumable and replayable.
- Pilot (10 cases per model): both models **diagnosed fewer cases correctly when gathering evidence themselves** than when given the full case; a **trace audit** tied part of the gap to two gatekeeper bugs, now being fixed. Next, with physician collaborators: a **"living patient" layer** (disease progression, treatment effects, test turnaround and Medicare-based costs, noisy tests, treat/wait/transfer actions) so agents can be **trained with RL**, not only evaluated.

Context hand-off in multi-stage coding agents (with PhD student Shivam Agarwal)

- Question: how much of what exploration sub-agents learn reaches the agent that writes the fix? Worked in the group's four-stage pipeline on OpenAI's **Codex CLI** (GPT-5.6) in Docker, explore (parallel sub-agents in separate git worktrees, emitting schema-checked JSON reports), synthesize, compact, solve, on **DeepSWE** (113 tasks) and **SWE-ReBench** (110).
- Designed and ran **six hand-off variants** on the base pipeline: compacted summary; fresh solver given the full report; a requirements brief that rules out re-exploration; a single-explorer ablation; findings injected into the solver's live session; a self-verification arm. Replicated an **oracle-context study** on Claude Code with Claude Sonnet 5 (**327 runs**).
- Built the evaluation: a 30-task dev set picked by leaderboard pass rates (**83 held out**), **paired significance tests**, and **log-forensics tooling** over Codex's JSONL logs (which prompt reached each stage, which model each sub-agent ran).
- Findings: **compaction discarded most of the exploration report** (median 11,500 → about 3,300 characters), and solvers resumed from an earlier session **nearly stopped spawning helper sub-agents** (2.5 per task → 0–0.4), the leading explanation for why two tuned pipelines solved **49 and 51** of the 83 held-out tasks versus **59 for the baseline**.
- Traced three tasks that failed even with the reference patch to **infrastructure faults** (a kernel inotify limit shared by root containers; a test-runner pool sized for 128 CPUs in a 2-CPU container); fixed the harness without touching graded tests, validated with reference and empty patches, and regraded saved patches (baseline **73** → **75** of 113 on DeepSWE).

UIUC AI Interpretability Group Urbana, IL
Undergraduate Researcher; mentor: Xiaocong Yang *Apr 2026 – present*

Do ReAct phase probes read an abstract representation? (sole author [2]; submitted to BlackboxNLP 2026)

- Asked whether Gemma-2 represents ReAct phase (Thought/Action/Observation) abstractly, and whether instruction tuning reshapes it. Built the stack (**TransformerLens**, **sparse autoencoders**, probing and ablation pipelines) and ran a **26-layer, 780-fit probe sweep** on Slurm over base and instruction-tuned Gemma-2 2B, then **replicated on 9B**.

- Instruction tuning **left phase decodability unchanged** (IT − base within ±0.002 across all 780 fits), falsifying the hypothesis that fine-tuning reshapes phase geometry. Phase stayed **about 98% decodable** at layer 12 with markers and newlines stripped, yet **28 of the 30** most phase-selective SAE features detect content type, and a position-decorrelation control showed the signal **decomposes into marker, content and position cues** rather than an abstract phase variable.

National Center for Supercomputing Applications (NCSA) Urbana, IL
SPIN Research Intern; mentors: Prof. Angela Lyons, Prof. Aiman Soliman *Aug 2025 – present*

Predicting internal migration in the Brazilian Amazon (six-student team)

- Where do people move, not just how many leave, as deforestation and warming intensify? Modeled **300K+ migration flows** between Brazil's **5,565 municipalities** with **135 features** each (census, climate, NDVI, agriculture).
- Built **graph neural networks** over a geographic-neighbor graph (never from the flows being predicted) with **distributed multi-GPU training on NCSA Delta**; the revised pipeline feeds GNN embeddings to **gradient-boosted trees**.
- Findings (correlational): the hottest quintile shows $\sim$**3**$\times$ **the out-migration** of the coolest; the poorest quintile shrinks (-2.2%), the richest grows ($+2.0\%$); SHAP ranks **agricultural employment** and **missing electricity and sewage** above income. Presented at the NCSA Students Research Conference (Apr 2026); extending to the **2022 Census**.

Internships & Consulting

Pocket FM San Francisco, CA
AI Research Intern; manager: Vasu Sharma, Head of AI *May 2026 – Aug 2026*

Listener-agent simulation (first-author paper [1], accepted at the Social Simulation Workshop, **COLM 2026**)

- After a summary-based persona agent scored near chance in a pilot (**AUC 0.558**) and lost to every simple non-LLM baseline, built a **baseline-first harness**: **1,199 tasks** across four frontier LLMs, grouped cross-fitting, bootstrap CIs.
- Findings: grounding agents in users' past actions **lowered Brier score for all four models** but raised ROC AUC for only two (+0.072, +0.058); a **calibrated non-LLM prior beat the grounded LLM hybrid** (Brier 0.216 vs. 0.235 on 306 tasks); adding digests of users' histories **made predictions worse** (Brier +0.013 on 150 tasks).
- Audited the engagement label the project relied on and found it **mixed exposure, skipping and missing telemetry** (11.3% of rows under 3 s of playtime); specified the **replacement event definition** used for downstream modeling.

Multimodal user simulation and production LLM systems

- Built **audio-visual user-simulation models** over **3,000+ shows and promos**, the company's first predictor of user response to visual promos (**63% accuracy**), developed with partner startups **Simile, Listen Labs and Kairosity**.
- Built **Adaptation**, the localization component of **Sherpa**, Pocket FM's AI writing platform (launched Sept 2026): it rewrites full scripts in another language and adapts names, settings and cultural references; **cut its cost by about 50%**.

Software Engineer Intern, Pocket FM *Bangalore, India May 2025 – Jul 2025*

- Built **"Blockbuster Engine"**, a launch-outcome predictor combining LoRA-tuned LLMs with gradient-boosted trees and neural networks: **80% accuracy vs. a 34% baseline**; after deployment, **unsuccessful content launches fell 40%**.

OTCR Consulting Urbana, IL
Technology Partner; founded the firm's technology department *Aug 2025 – present*

- Lead **21 technology consultants** on internal software (operations dashboard, recruiting and assessment platforms, RAG knowledge service), client projects (a healthcare-AI startup; an enterprise agentic-AI program) and technical trainings.

Projects & Ventures

Phantom, autonomous web penetration-testing agent *HackIllinois 2026;* ***YC Startup School 2026***

- A Playwright crawler maps a target's attack surface; a **ReAct loop** plans and executes **multi-step exploit chains** (SQL injection, XSS, IDOR, CSRF, authentication bypass) and returns **CVSS-scored findings**.
- Selected for **Y Combinator's Startup School** in San Francisco (July 2026); received $25K in compute credits from YC partners (OpenAI, Anthropic, AWS, Azure).

BeClear, AI college-admissions advising agent *2024 – 2025*

- RAG-based advising agent with LoRA-fine-tuned models; **200+ users across 3 partner schools**; **acquired in 2025**.

Honors & Skills

Honors: NCSA Letter of Recognition from Director R. Srikant for outstanding contributions to a SPIN project, nominated by mentor Prof. Angela Lyons (2026); Winner (1st of 2,000+), New York Academy of Sciences Global Challenge (2023).

Skills: Python, C++, SQL, R, TypeScript; PyTorch, Hugging Face, TransformerLens, vLLM; Docker, Slurm, Kubernetes, AWS.